Characteristics of White Leghorn Chickens Selected for Heat Tolerance[1]

H. R. WILSON, C. J. WILCOX,[2] R. A. VOITLE, C. D. BAIRD[3] AND R. W. DORMINEY[4]
Department of Poultry Science, Florida Agricultural Experiment Station, Gainesville, Florida 32611

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ABSTRACT A series of studies was conducted to identify and delineate factors associated with responses of chickens to high ambient temperatures. Two populations of birds were used which had been selected divergently for tolerance to stress at 40.6° C. and 75% R.H. and at 5 weeks of age.

After four generations of selection for survival time the lines had separated by 28 minutes: 58 *vs.* 86 minutes for the low and high, respectively. Egg production, hatchability, Haugh units and specific gravity did not differ between lines. Egg weight was greater in the low line.

The body temperature of chicks at 30 minutes of heat stress was correlated ($r = -0.454$; $P < 0.05$) to survival time. Body weight was correlated ($r = -0.325$) with resistance to heat stress while loss of body weight during heat stress was correlated ($r = +0.478$) with resistance. There was no apparent difference between lines in respect to resistance of 16-day old embryos to heat stress.

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INTRODUCTION

THE ability of farm animals to produce efficiently under unfavorable environmental conditions, particularly high temperature and/or humidity, is very important economically. This characteristic appears to be extremely complex physiologically and it is influenced by numerous environmental and genetic factors.

Chickens respond to unfavorable environments with alterations in body temperature, respiration rate, water consumption, egg production and mortality. Response differences have been noted between breeds (Hutt, 1938; Yeates *et al.*, 1941; Lee *et al.*, 1945) and families within breeds (KeirEldin and Shaffner, 1954; Wilson *et al.*, 1966). Heritability of survival time for chicks exposed to high temperature was estimated by Wilson *et al.* (1966) to be between 0.29 and 0.31.

Objectives of the present studies were to identify and delineate environmental and genetic factors affecting responses of chickens to high ambient temperatures.

PROCEDURE

Two populations of chicks were used in these studies which had been selected divergently for heat tolerance (Wilson *et al.*, 1966). Chicks were tested for heat tolerance (30 per run) at five weeks of age in a heat chamber previously described (Wilson *et al.*, 1966) at 40.8 ± .3° C., 0.7079 $m.^3$ per minute air flow rate and 75% relative humidity, and individual survival time recorded in minutes. Chamber temperature was recorded and found to vary slightly between runs and decrease approximately 1.9° C. during placement of chicks in the chamber.

The chicks were grown under infra-red brooding lamps and fed a commercial-type starter diet *ad lib.* until the time of heat exposure. Neither feed nor water was available during exposure. Exposure time was a maximum of 120 minutes and any chicks alive at this point were assigned a survival time of 120 minutes.

1. Florida Agr. Exp. Sta. Journal Series No. 5377.

2. Department of Dairy Science, University of Florida, Gainesville, Florida.

3. Department of Agricultural Engineering, University of Florida, Gainesville, Florida.

4. Department of Poultry Science, Oregon State University, Corvallis, Oregon.

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Genetic Aspects. Based on response of 902 progeny of the base population (42 sires and 147 dams), outstanding sires (upper and lower 19%) and dams (upper and lower 30%) were selected for further matings to produce divergent lines, high and low tolerance, representing survival times. Offspring of these matings were mated and their offspring tested for survival time.

Data were analyzed by the method of least-squares (Harvey, 1960) for 1) the original base population, and 2) the high and low lines. Estimates of environmental effects were assumed to be represented by the least-squares constants. Evidence of genetic effects arose from estimates of heritability from the base population using paternal half-sib correlation (Falconer, 1960) and the separation of the survival means in the high and low lines. The base population and four subsequent generations were tested for heat tolerance.

Performance Characteristics. Egg production and hatchability were obtained for the F_1, F_2 and F_3 generations. Hatchability was determined when chicks were hatched for evaluating each generation. Egg quality was determined at 200 and 365 days of age for the F_1 birds. Adult birds were maintained in floor pens in an open type house with one male and approximately 10 hens per pen.

Body Temperature. Two experiments were conducted in which the relationship between survival and body temperatures were examined. In the first experiment 32 low line and 29 high line chicks were heat stressed and body temperature measured after 30 minutes of exposure. The correlation of body temperature with subsequent survival time was calculated. In the second experiment 177 low line and 124 high line chicks were heat stressed and body temperature measured throughout the exposure period. Body temperature was measured with a 24-point recording potentiometer with the thermocouple placed subcutaneously on the body wall underneath the thigh. Although body position can influence body wall temperature, the location of the probe and close confinement of the birds minimized this variable.

Weight Loss. A total of 287 chicks (144 low line and 143 high line) in two experiments hatched from the F_3 adults were weighed when placed in the heat chamber and again when removed from the chamber. Correlation coefficients were calculated for survival time and percent body weight loss and for survival time and initial body weight.

Heat Stress of Embryos. Three experiments were conducted to determine if embryos of the low and high lines would react to heat stress in a similar fashion to the five-week old chicks. In the first experiment approximately 60 embryos, 16 days of age, from each line were exposed to 2 hours of 40.6° C. and 62% relative humidity in a Jamesway 252 incubator, while another 60 embryos were exposed at those conditions for 6 hours. In the second experiment approximately 140 and 220 embryos (16-day) from the low and high lines, respectively, were divided into two groups per line and exposed to 40.6° C. and 62% relative humidity for 6 or 24 hours in the incubator. A total of 453 and 434 embryos from the low and high lines, respectively, were randomly divided into five groups and

TABLE 1.—*Survival time of five generations of chicks heat stressed at 40.8° C. (105° F.) and 75% relative humidity*

Generation	Survival time (min.)[1]	
	Low line	High line
F_0	74	—
F_1	48[a]	61[b]
F_2	63[a]	86[b]
F_3	62[a]	84[b]
F_4	58[a]	86[b]

[1] Means within each generation having different superscripts are significantly different ($P < 0.05$).

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exposed to one of the following treatments in experiment 3: control (37.5° C.); 40.6° C. for 6 or 24 hours; 43.3° C. for 6 or 24 hours.

After exposure all embryos were returned to the control incubator and held at normal incubation conditions until hatch. The effect of heat stress was measured in terms of hatchability of fertile eggs.

RESULTS AND DISCUSSION

Genetic Aspects. Average survival time for the base population, as previously reported (Wilson *et al.*, 1966), was 74 minutes. The high and low lines, after one generation of selection, differed by 13 minutes (Table 1) with average survival times being 61 and 48 minutes, respectively. When adjusted for sex, temperature and age by the use of the least-squares constants for these effects, adjusted means were 59 minutes for the high line and 48 minutes for the low line, a significant difference ($P < 0.01$). Means of both F_1 lines were below the overall mean for the base population due to the removal of a small circulation fan from the test chamber, but returned to the original level in the F_2 generation when the fan was replaced. Heritability of survival time was estimated to be 0.291 ± 0.096. Further separation of the lines was obtained in the F_2 and F_4 generations to a maximum difference of 28 minutes.

Performance Characteristics. Egg production and hatchability for three generations showed no significant differences between high and low lines (Table 2). There is no explanation for the wide variation from year to year. Egg weight was greater ($P < 0.05$) in the low line (Table 3) at both 200 and 365 days of age. Interior quality (Haugh units) and specific gravity did not differ significantly between lines (Table 3).

Body Temperature. When body temperature after 30 minutes of heat exposure was tested against subsequent survival time in Experiment 1 a significant ($P < 0.05$) relationship was observed ($r = -0.294$). The correlation was higher in the second experiment ($r = -0.573$; $P < 0.05$) and was −0.454 for the two experiments combined.

Body temperature tended to reach higher end points in the low line birds (Fig. 1) than in the high line birds (Fig. 2). A temperature-time relationship exists within the high line, as well as between lines, in which the birds

TABLE 2.—*Egg production and hatchability (of all eggs set) of birds genetically selected for resistance to heat stress*

Generation	Egg prod. (% HD)		Hatchability (%)	
	Low line	High line	Low line	High line
F_1	61	58	80	83
F_2	47	41	35	41
F_3	67	73	72	66

TABLE 3.—*Weight, Haugh units and specific gravity of eggs from birds selected for resistance to heat stress*

	200 Days of Age[1]		365 Days of Age[1]	
	Low line	High line	Low line	High line
Egg wt. (gm.)	47.3[a]	45.8[b]	56.5[a]	54.3[b]
Haugh units	88.6	90.3	72.2	71.4
Specific gravity	1.091	1.091	1.081	1.080

[1]Means within an age period with different superscripts are significantly different ($P < 0.05$).

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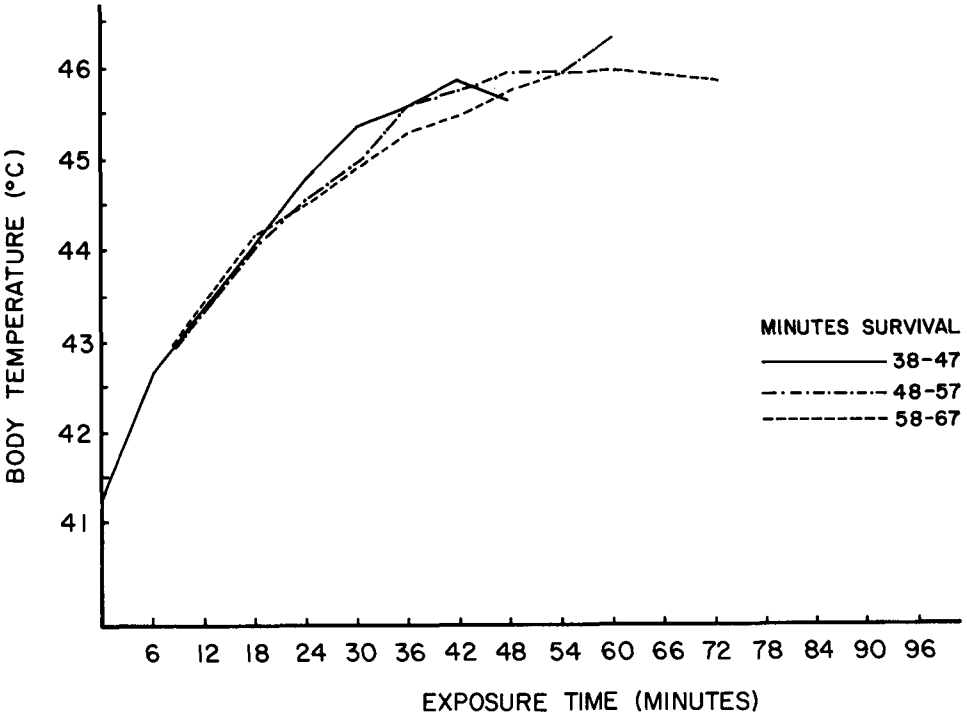


FIG. 1. Body temperature of Low Line birds during heat stress and grouped according to survival time.

that live longer tend to reach a lower terminal body temperature. This would indicate that the tolerance of these birds to heat is associated more with an ability to prevent body temperature increases than to an ability to withstand higher body temperatures.

Weight Loss. The birds exhibiting greater tolerance to heat stress were found to lose a greater percent of their body weight during stressing ($r = +.478$; $P < 0.05$). Weight loss was 7.8% for high line and 6.9% for low line chicks. Birds which live longer would be expected to experience greater weight loss due in part to the longer period of active respiratory evaporation. The rate of evaporative loss was not determined but could have differed between birds. A significant ($P < 0.05$) negative correlation ($r = -0.325$) existed between initial body weight and resistance to heat stress.

Heat Stress of Embryos. The exposure of 16-day old embryos to 40.6° C. for up to 24 hours essentially had no effect on their hatchability, regardless of heat tolerance line (Table 4). Exposure at 43.3° C. for 6 hours (Exp. 3) caused a slight decrease in hatchability in both lines. However, 24 hours at 43.3° C. killed all embryos of both lines. Therefore, no major differences appear to exist between lines in respect to embryo

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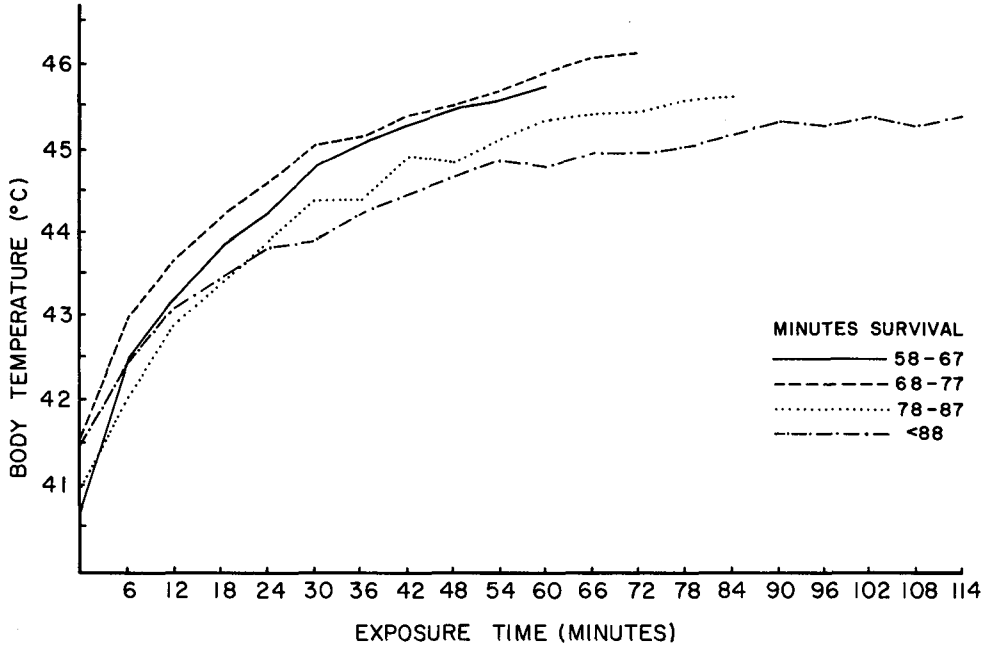


FIG. 2. Body temperature of High Line birds during heat stress and grouped according to survival time.

TABLE 4.—*Hatchability of embryos heat stressed at 16 days of age*

Stress	% Hatchability	
	Low line	High line
	Exp. 1	
2 hrs., 40.6° C.	86	93
6 hrs., 40.6° C.	92	95
	Exp. 2	
Cont., 37.5° C.	99	95
6 hrs., 40.6° C.	97	94
24 hrs., 40.6° C.	91	92
	Exp. 3	
Cont., 37.5° C.	94	96
6 hrs., 40.6° C.	99	92
24 hrs., 40.6° C.	91	94
6 hrs., 43.3° C.	87	91
24 hrs., 43.3° C.	0	0

susceptibility to heat stress indicating that the physiological mechanism causing line separation at five weeks of age is not effective in the 16-day old embryo.

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